Fiscal Unions

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Roadmap

- Motivation and the two key results
- Static model: environment and the labor wedge
- The aggregate-demand externality
- Optimal policy and implementation
- Dynamic NK model and welfare loss
- Optimal transfers in closed form
- Numerical illustration
- Takeaways

Why do currency unions need fiscal unions?

The fundamental problem. Friedman (1953): flexible exchange rates are valuable for adjusting to asymmetric shocks. Currency-union members give them up.

The optimal currency area (OCA) literature.

- Mundell (1961): factor mobility
- McKinnon (1963): openness lowers costs
- **Kenen (1969): fiscal integration**

Eurozone crisis made the question concrete: asymmetric shocks (Spain vs. Germany), no exchange-rate adjustment, no systematic transfers $\Rightarrow$ deep peripheral recessions.

This paper. How should a fiscal union be designed, and how effective can it be?

Result 1 – Currency-union members value risk sharing more

Under **incomplete markets**, the value of any given level of risk sharing is *strictly greater* for a currency-union member than for a country with a flexible exchange rate.

Why? The country's indirect utility under a fixed exchange rate lies below the upper envelope it could attain by adjusting p:

$$V^i(C_T, p; s) \;\leq\; \max_p V^i(C_T, p; s) \;\equiv\; V^{i*}(C_T; s),$$

with equality only when $\tau_i(s) = 0$.

Hence V^i is more concave than V^{i*} in C_T: members are *more risk averse*, so insurance is worth more.

Result 2 – Private risk sharing is constrained inefficient

Under **complete markets**, privately optimal risk sharing is *constrained Pareto inefficient* inside a currency union.

Why? The private and social marginal values of a transfer differ:

$$\underbrace{V^i_{C_T}(s)}_{\text{social}} = \underbrace{U^i_{C_T}(s)}_{\text{private}}\left[1 + \frac{\alpha_i(s)}{p_i(s)}\,\tau_i(s)\right].$$

Private agents do not internalize the bracketed correction – the *aggregate-demand externality*.

⇒ A fiscal union is needed even when financial markets are complete.

Static model: environment

Setup.

- Continuum of countries $i \in [0, 1]$, single period, state $s \in S$
- Density $\pi(s)$ over states

Two commodities.

- Traded good with endowment $E_i^T(s)$ (law of one price)
- Nontraded good $Y_{i,j}^{NT}(s) = A_i(s)N_{i,j}(s)$ (CES, monopolistic firms)

Two key frictions.

- *Common currency* forces $P_T(s)$ to be the same across countries.
- *Sticky prices* force P_i^{NT} to be set before s is realized.

The central friction

The relative price of traded to nontraded goods,

$$p_i(s) \equiv \frac{P_T(s)}{P_i^{NT}},$$

must factorize as a (function of s) over a (function of i).

$P_T(s)$ varies with state s only	$\div$	P_i^{NT} varies with country i only	$=$	$p_i(s)$ cannot vary freely

- Flexible exchange rate: $P_i^T(s) = E_i(s)P_T(s)$ adjusts freely $\Rightarrow$ first best.
- Currency union: factorization binds $\Rightarrow$ first best generically unattainable.

Households: preferences and budget constraints

Preferences. Household i maximizes expected utility

$$\int_S U^i\left(C_i^{NT}(s),\, C_i^T(s),\, N_i(s);\, s\right)\pi(s)\,ds.$$

Portfolio budget (date 0). $D_i(s)$: nominal payoff in state s; $Q(s)$: Arrow–Debreu price.

$$\int_S D_i(s)\,Q(s)\,\pi(s)\,ds \;\leq\; 0.$$

Flow budget (date 1, in state s).

$$P_i^{NT}C_i^{NT}(s) + P_T(s)C_i^T(s) \;\leq\; W_i(s)N_i(s) + P_T(s)E_i^T(s) + \Pi_i(s) + T_i(s) + (1+\tau_i^D(s))D_i(s).$$

W_i: nominal wage. Π_i: profits. T_i: lump-sum rebate. $\tau_i^D(s)$: state-contingent portfolio return subsidy.

Households: Lagrangian

Multiplier Λ_i on portfolio budget; multiplier $\mu_i(s)\pi(s)$ on flow budget. Form

$$\mathcal{L}_i \;=\; \int_S \Big[U^i(s) + \mu_i(s)\,(\text{flow budget gap})\Big]\,\pi(s)\,ds \;-\; \Lambda_i \int_S D_i(s)Q(s)\pi(s)\,ds.$$

Differentiate with respect to $D_i(s)$:

$$\mu_i(s)\,(1+\tau_i^D(s)) \;=\; \Lambda_i\,Q(s) \quad\Longrightarrow\quad \mu_i(s) \;=\; \frac{\Lambda_i\,Q(s)}{1+\tau_i^D(s)}.$$

Differentiate with respect to $C_i^T(s)$, $C_i^{NT}(s)$, $N_i(s)$:

$$\begin{aligned} U^i_{C_T}(s) &= \mu_i(s)\,P_T(s), \\ U^i_{C_{NT}}(s) &= \mu_i(s)\,P_i^{NT}, \\ -U^i_N(s) &= \mu_i(s)\,W_i(s). \end{aligned}$$

Households: first-order conditions

Combining the four FOCs and eliminating μ_i:

$$\frac{U^i_{C_T}(s)\,[1+\tau^D_i(s)]}{Q(s)\,P_T(s)} = \text{const across } s \qquad \text{(Euler)}$$

$$\boxed{\frac{U^i_{C_T}(s)}{U^i_{C_{NT}}(s)} = \frac{P_T(s)}{P^{NT}_i} = p_i(s)} \qquad \text{(intratemporal)}$$

$$-\frac{U^i_N(s)}{U^i_{C_{NT}}(s)} = \frac{W_i(s)}{P^{NT}_i} \qquad \text{(labor supply)}$$

Firms: technology and CES demand

CES aggregator over varieties $j \in [0,1]$:

$$Y_i^{NT}(s) = \left[\int_0^1 Y_{i,j}^{NT}(s)^{1-1/\varepsilon}\, dj\right]^{\varepsilon/(\varepsilon-1)}, \qquad \varepsilon > 1.$$

Demand for variety j:

$$Y_{i,j}^{NT}(s) = C_i^{NT}(s)\left(\frac{P_{i,j}^{NT}}{P_i^{NT}}\right)^{-\varepsilon}, \qquad P_i^{NT} = \left[\int_0^1 (P_{i,j}^{NT})^{1-\varepsilon} dj\right]^{1/(1-\varepsilon)}.$$

Production: $Y_{i,j}^{NT}(s) = A_i(s)\, N_{i,j}(s)$. Wage paid by the firm: $(1+\tau_i^L)\, W_i(s)$.

Sticky prices. Each monopolist must set $P_{i,j}^{NT}$ *before* s is realized $\Rightarrow$ price independent of s.

Firms: optimal price setting

Stochastic discount factor used by the firm: $Q(s)/(1+\tau_i^D(s))$ (matches the household SDF).

Variety j chooses $P_{i,j}^{NT}$ to maximize

$$\int_S \frac{Q(s)}{1+\tau_i^D(s)}\left[P_{i,j}^{NT} - \frac{(1+\tau_i^L)W_i(s)}{A_i(s)}\right] C_i^{NT}(s)\left(\frac{P_{i,j}^{NT}}{P_i^{NT}}\right)^{-\varepsilon} \pi(s)\,ds.$$

At a symmetric equilibrium ($P_{i,j}^{NT} = P_i^{NT}$), the FOC delivers

$$P_i^{NT} = (1+\tau_i^L)\,\frac{\varepsilon}{\varepsilon-1}\,\frac{\int_S \frac{Q(s)}{1+\tau_i^D(s)}\frac{W_i(s)}{A_i(s)}\,C_i^{NT}(s)\,\pi(s)\,ds}{\int_S \frac{Q(s)}{1+\tau_i^D(s)}\,C_i^{NT}(s)\,\pi(s)\,ds} \doteq (1+\tau_i^L)\,\frac{\varepsilon}{\varepsilon-1}\,\mathbb{E}_{\text{supp}}[W_i(s)/A_i(s)].$$

A markup $\varepsilon/(\varepsilon-1)$ over an SDF-weighted average real marginal cost.

Government and union resource constraints

Country i government budget.

$$T_i(s) \;=\; \tau_i^L\, W_i(s)\, N_i(s) \;-\; \tau_i^D(s)\, D_i(s) \;+\; \hat{T}_i(s).$$

Union-wide feasibility for transfers.

$$\int_I \hat{T}_i(s)\, di \;=\; 0 \qquad \forall\, s \in S.$$

Policy instruments at the planner's disposal.

- Labor tax/subsidy τ_i^L (offsets the monopoly markup and pins down P_i^{NT} via (6)).
- State-contingent portfolio tax $\tau_i^D(s)$ (only useful with complete markets).
- State-contingent international transfers $\hat{T}_i(s)$ (the focus of the paper).

Equilibrium with complete markets

Market clearing.

$$
\begin{aligned}
C_i^{NT}(s) &= A_i(s)\, N_i(s), \\
\int_I C_i^T(s)\, di &= \int_I E_i^T(s)\, di, \\
\int_I D_i(s)\, di &= 0 \qquad \forall\, s.
\end{aligned}
$$

Equilibrium.
An allocation of $\{C_i^{NT}, C_i^T, N_i, D_i\}$, $\{P_i^{NT}, P_T(s), W_i(s), Q(s)\}$ given taxes $\{\tau_i^L, \tau_i^D(s)\}$ and transfers $\{\hat{T}_i(s)\}$.

Equilibrium with incomplete markets

No private financial markets $(D_i \equiv 0)$. Households face the sequential budget constraint

$$P_i^{NT} C_i^{NT}(s) + P_T(s) C_i^T(s) \;=\; W_i(s) N_i(s) + P_T(s) E_i^T(s) + \Pi_i(s) + T_i(s).$$

Firms use the household SDF $U^i_{C_T}(s)/P_T(s)$ directly:

$$P_i^{NT} \;=\; (1+\tau_i^L)\,\frac{\varepsilon}{\varepsilon-1}\,\frac{\int_S U^i_{C_T}(s)\, W_i(s)/[A_i(s) P_T(s)]\, C_i^{NT}(s)\, \pi(s)\, ds}{\int_S U^i_{C_T}(s)/P_T(s)\, C_i^{NT}(s)\, \pi(s)\, ds}.$$

Government budget simplifies to

$$T_i(s) \;=\; \tau_i^L\, W_i(s)\, N_i(s) \;+\; \hat{T}_i(s).$$

Implementability: same set under both market structures

Proposition (1–2)

An allocation $\{C_i^T(s), C_i^{NT}(s), N_i(s)\}$ *with prices* $\{P_T(s), P_i^{NT}\}$ *is part of an equilibrium iff*

$$\frac{U^i_{C_T}}{U^i_{C_{NT}}} = \frac{P_T(s)}{P_i^{NT}}, \qquad C_i^{NT} = A_i N_i, \qquad \int_I C_i^T\, di = \int_I E_i^T\, di$$

hold. Same set under complete and incomplete markets.

Implementability constraint is the only constraint that genuinely restricts allocations.

What changes between the two market structures

Same allocation set, *different* policy instruments.

Complete markets ($D_i(s)$ traded on AD markets)

Instruments: $\tau_i^L + \tau_i^D(s) + T_i(s)$ (labor tax + portfolio tax + lump-sum).

- $\tau_i^D(s)$ corrects state-contingent insurance.
- $\hat{T}_i(s)$ *indeterminate*: markets vs. government on a continuum.

Incomplete markets ($D_i \equiv 0$)

Instruments: $\tau_i^L + \hat{T}_i(s)$ (labor tax + state-contingent transfer).

- $\hat{T}_i(s)$ does *all* the cross-country resource shifting: $\hat{T}_i(s) = P_T(s)(C_i^T(s) - E_i^T(s))$.
- No portfolios to distort $\Rightarrow$ no portfolio taxes.

Homothetic preferences and the relative price

Assumptions.

- Preferences over (C^T, C^{NT}) are weakly separable from labor.
- Preferences over (C^T, C^{NT}) are homothetic.

Implication. The ratio of nontraded to traded consumption depends only on relative prices:

$$C_i^{NT}(s) \;=\; \alpha^i\,(p_i(s);\,s)\;C_i^T(s), \qquad \alpha_p^i(\cdot;s) > 0.$$

This single function $\alpha^i(p;s)$ encapsulates implementability constraint:

$$\frac{U^i_{C_T}\left(\alpha^i(p;s)C_T,\, C_T,\cdots\right)}{U^i_{C_{NT}}\left(\alpha^i(p;s)C_T,\, C_T,\cdots\right)} \;=\; p \quad\Longleftrightarrow\quad C^{NT} = \alpha^i(p;s)C_T.$$

First-best benchmark

A first-best allocation requires:

1. Static efficiency (no labor wedge):

$$-\frac{U^i_N(s)}{U^i_{C_{NT}}(s)} = A_i(s) \qquad \forall\, i, s \quad \iff \quad \tau_i(s) = 0.$$

2. Optimal risk sharing (Borch rule):

$$\frac{U^i_{C_T}(s)}{U^i_{C_T}(s')} = \frac{U^{i'}_{C_T}(s)}{U^{i'}_{C_T}(s')} \qquad \forall\, i, i', s, s'.$$

Generically incompatible with eq. (4) under sticky prices and a common currency $\Rightarrow$ first best unattainable in a currency union; we look for the second best.

The labor wedge

Definition

$$\tau_i(s) \;\equiv\; 1 + \frac{U^i_N(s)}{A_i(s)\, U^i_{C_{NT}}(s)} = \frac{p_i^{NT} - W_i(s)/A_i(s)}{p_i^{NT}}$$

- Recall that $p_i^{NT} = \mathbb{E}_{\text{supp}}[W_i(s)/A_i(s)]$.
- $\tau_i(s) = 0$ (**first best**): $p_i^{NT} = W_i(s)/A_i(s)$ and $-U_N/U_{C_{NT}} = A(s)$.
- $\tau_i(s) > 0$ (**recession**): $p_i^{NT} > W_i(s)/A_i(s)$ output below potential (demand too weak).
- $\tau_i(s) < 0$ (**boom**): $p_i^{NT} < W_i(s)/A_i(s)$ output above potential (demand too strong; overheating).

The labor wedge is the central diagnostic statistic and appears in every optimal-policy formula below.

Indirect utility

Under separability and homotheticity, $C_i^{NT}(s) = \alpha^i(p;s)\, C_i^T(s)$. Define

$$V^i(C_T, p; s) \;=\; U^i\left(\alpha^i(p;s)C_T,\, C_T,\, \tfrac{\alpha^i(p;s)}{A_i(s)}C_T;\, s\right).$$

Proposition 3

$$V_p^i(s) \;=\; \frac{\alpha_p^i(s)}{p_i(s)}\, C_i^T(s)\, U_{C_T}^i(s)\, \tau_i(s),$$

$$V_{C_T}^i(s) \;=\; U_{C_T}^i(s)\left[1 + \frac{\alpha_i(s)}{p_i(s)}\,\tau_i(s)\right].$$

The aggregate-demand externality

The wedge between social and private values of a transfer:

$$V^i_{C_T}(s) - U^i_{C_T}(s) = U^i_{C_T}(s)\,\frac{\alpha_i(s)}{p_i(s)}\,\tau_i(s) = U^i_{C_T}(s)\,\frac{P_i^{NT}C_i^{NT}}{P_T C_i^T}\,\tau_i(s).$$

- Recession ($\tau > 0$): private agents *undervalue* transfers.
- Boom ($\tau < 0$): private agents *overvalue* the cost of transfers.
- Wedge weighted by relative nontraded expenditure share.

Private agents do not internalize that their insurance choices shift aggregate demand and so output and employment in other countries.

Keynesian multiplier

- Country in recession ($\tau > 0$) receives a transfer.
- Households spend on *both* traded and nontraded goods.
- P_i^{NT} is sticky $\Rightarrow$ the demand increase translates one-for-one into nontraded *output*.
- More output $\Rightarrow$ more income $\Rightarrow$ more spending: a Keynesian multiplier.

This is the mechanism behind the famous *Transfer Problem* (Keynes 1929 vs. Ohlin 1929). Under sticky prices and home bias, transfers have real macroeconomic effects – not just consumption-smoothing effects.

The Ramsey planning problem

Second-best problem

$$\max_{P_T(s),\, P_i^{NT},\, C_i^T(s)} \int_I \int_S \lambda^i V^i\left(C_i^T(s), \frac{P_T(s)}{P_i^{NT}}; s\right) \pi(s)\, ds\, di$$

s.t. $\int_I C_i^T(s)\, di = \int_I E_i^T(s)\, di.$

Let $\mu(s)\pi(s)$ be the multiplier on the resource constraint. The FOCs are

$$\begin{aligned} [\, C_i^T\,] : \lambda^i V^i_{C_T}(s) &= \mu(s), \\ [\, P_T(s)\,] : \int_I V^i_p(s)\, \frac{\lambda^i}{P_i^{NT}}\, di &= 0, \\ [\, P_i^{NT}\,] : \int_S V^i_p(s)\, p_i(s)\, \lambda^i\, \pi(s)\, ds &= 0. \end{aligned}$$

Optimal price setting

Substituting Prop. 3 into the FOC for P_i^{NT}:

$$\int_S \alpha_p^i(s)\, C_i^T(s)\, U_{C_T}^i(s)\, \tau_i(s)\, \pi(s)\, ds \;=\; 0.$$

- For each country i, the *weighted average labor wedge across states* equals zero.
- Implemented by the labor tax τ_i^L, which pins down P_i^{NT}.
- Without uncertainty, this collapses to $\tau_i = 0$ (the first best).

Optimal monetary policy

Substituting Prop. 3 into the FOC for $P_T(s)$:

$$\int_I \alpha_p^i(s)\, C_i^T(s)\, U_{C_T}^i(s)\, \tau_i(s)\, \lambda^i\, di \;=\; 0.$$

- For each state s, the *weighted average labor wedge across countries* equals zero.
- Union-wide monetary policy targets the “average” economy.
- Symmetric shocks $\Rightarrow$ $\tau_i(s) = 0$ for all i (perfect stabilization).
- Asymmetric shocks $\Rightarrow$ heterogeneous wedges unavoidable.

Constrained efficient risk sharing

From the FOC for C_i^T, $\lambda^i V^i_{C_T}(s) = \mu(s)$. Taking ratios across states and countries:

$$\frac{V^i_{C_T}(s)}{V^i_{C_T}(s')} = \frac{V^{i'}_{C_T}(s)}{V^{i'}_{C_T}(s')}, \qquad V^i_{C_T}(s) = U^i_{C_T}(s)\left[1 + \frac{\alpha_i(s)}{p_i(s)}\,\tau_i(s)\right].$$

This is the Borch rule applied to *social* marginal values rather than to private marginal utilities.

Contrast with privately optimal risk sharing:

$$\frac{U^i_{C_T}(s)}{U^i_{C_T}(s')} = \frac{U^{i'}_{C_T}(s)}{U^{i'}_{C_T}(s')}.$$

Inefficiency of private risk sharing

Proposition (7)

Under complete markets without portfolio taxes, the competitive equilibrium is constrained Pareto inefficient unless $\tau_i(s) = 0$ for all (i, s), in which case it is first best.

Sketch. For the constrained-efficient and private conditions to hold simultaneously, the ratio $[1 + (\alpha_i/p_i)\tau_i(s)]/[1 + (\alpha_i/p_i)\tau_i(s')]$ must be the same constant for all i.

But Props. 4–5 require weighted averages of τ across states and countries to be zero. These conditions are generically incompatible unless $\tau \equiv 0$.

$\Rightarrow$ Even with complete markets, government intervention in risk sharing is needed.

Implementation under complete markets

Implement the constrained efficient allocation with state-contingent portfolio taxes

$$\tau_i^D(s) = \frac{\alpha_i(s)}{p_i(s)} \tau_i(s)$$

- **Subsidize** insurance for bad states ($\tau > 0$).
- **Tax** insurance for good states ($\tau < 0$).
- Independent of Pareto weights λ^i – corrects an externality, does not redistribute.

Implementation under incomplete markets

Implement the same allocation with state-contingent transfers

$$\hat{T}_i(s) \;=\; P_T(s)\,(C_i^T(s) - E_i^T(s))$$

- Government finances the entire current-account gap.
- Households have no assets to undo transfers $\Rightarrow$ no portfolio restrictions are needed.
- These transfers *go beyond* replicating the complete-markets outcome – because the latter is itself constrained inefficient.

Countries outside the currency union

A country with a flexible exchange rate sets $P_i^T(s) = E_i(s)P_T(s)$ freely. The FOC becomes $V_p^i = 0$, hence

$$\tau_i(s) = 0 \qquad \forall s \in S.$$

- No labor wedge $\Rightarrow$ no aggregate-demand externality.
- Risk-sharing condition (15) collapses to (16).
- Complete markets: no portfolio taxes needed.
- Incomplete markets: optimal transfers *replicate* the complete-markets outcome (do not improve upon it).

The case for fiscal unions is specific to currency unions.

Dynamic NK model: setup

Households.

$$\sum_{t=0}^{\infty} \beta^t \left[\frac{C_t^{1-\sigma}}{1-\sigma} - \frac{N_t^{1+\varphi}}{1+\varphi} \right], \qquad C_t = \left[(1-\alpha)^{\frac{1}{\eta}} C_{H,t}^{\frac{\eta-1}{\eta}} + \alpha^{\frac{1}{\eta}} C_{F,t}^{\frac{\eta-1}{\eta}} \right]^{\frac{\eta}{\eta-1}}.$$

Firms. $Y_t(j) = A_{H,t} N_t(j)$; Calvo pricing (fraction $1-\delta$ resets each period); $\tau^L = -1/\varepsilon$ offsets the markup.

Country budget. $\widehat{NFA}_0^i = (\alpha/\rho)\,\bar{\theta}^i$ – the transfer is an endogenous initial NFA position.

Cole–Obstfeld benchmark and notation

Cole–Obstfeld case: $\sigma = \eta = \gamma = 1$.

- Log utility, unit elasticities.
- Trade balanced under no transfers (Backus–Smith).
- Any $\bar{\theta}^i \neq 0$ at the optimum signals that complete markets would be constrained inefficient.
- Yields a clean second-order welfare approximation.

Notation (log deviations from symmetric s.s.).

- $\hat{y}^i_t$, $\hat{\pi}^i_{H,t}$: country i output gap, PPI inflation gap.
- Star = union average; bars = deviations from union average.
- $\tilde{s}^i_t = a_{i,t} - a^*_t$: idiosyncratic shock.
- $\bar{\theta}^i$: time-invariant transfer parameter.

Log-linearized system in gaps

Disaggregated block (each country i).

$$\begin{aligned}
\dot{\bar{\pi}}^i_{H,t} &= \rho\,\bar{\pi}^i_{H,t} - \kappa_y\,\bar{y}^i_t - \lambda\alpha\,\bar{\theta}^i, \\
\dot{\bar{y}}^i_t &= -\bar{\pi}^i_{H,t} - \dot{\tilde{s}}^i_t, \\
\bar{y}^i_0 &= (1-\alpha)\,\bar{\theta}^i - \tilde{s}^i_0, \\
\int_0^1 \bar{\theta}^i\,di &= 0,
\end{aligned}$$

with $\lambda = \rho_\delta(\rho + \rho_\delta)$ and $\kappa_y = \lambda(1+\varphi)$.

Aggregate block.

$$\dot{\pi}^*_t = \rho\,\pi^*_t - \kappa_y\,\hat{y}^*_t.$$

Welfare loss function

Second-order approximation to utilitarian welfare around the symmetric steady state (cf. Farhi–Werning 2012):

$$\mathcal{L} = \frac{1+\varphi}{2} \int_0^\infty \int_0^1 e^{-\rho t} \left[\alpha_\pi (\hat{\pi}^i_{H,t})^2 + (\hat{y}^i_t)^2 + \alpha_\theta (\bar{\theta}^i)^2 \right] di\, dt,$$

with $\alpha_\pi = \dfrac{\varepsilon}{\lambda(1+\varphi)}$ and $\alpha_\theta = \dfrac{\alpha(2-\alpha)}{1+\varphi}$.

- $\alpha_\pi \hat{\pi}^2$: standard Calvo cost of price dispersion.
- $\hat{y}^2$: standard cost of demand fluctuations.
- $\alpha_\theta \bar{\theta}^2$: deviation from privately optimal risk sharing; vanishes as $\alpha \to 0$.

Aggregate and disaggregated planning problems *decouple*.

Optimal monetary policy

The aggregate planner solves

$$\min \frac{1}{2}\int_0^\infty e^{-\rho t}\left[\alpha_\pi(\pi_t^*)^2 + (\hat{y}_t^*)^2\right]dt \quad \text{s.t.} \quad \dot{\pi}^* = \rho\pi^* - \kappa_y \hat{y}^*.$$

Proposition (13)

At the second-best optimum, $\hat{y}_t^* = \pi_t^* = 0$ for all $t \geq 0$.

- Union policy perfectly stabilizes the symmetric component.
- Powerless against asymmetric shocks: a single rate cannot stabilize Spain and Germany simultaneously.
- Hence the role for fiscal transfers $\bar{\theta}^i$.

Optimal transfers under rigid prices

Set $\kappa_y = 0$. Inflation is pinned at zero by (23); (24) integrates to $\bar{y}^i_t = (1-\alpha)\bar{\theta}^i - \tilde{s}^i_t$.

Country i's sub-problem reduces to

$$\min_{\bar{\theta}^i} \; \tfrac{1}{2}\int_0^\infty e^{-\rho t}\Big[\big((1-\alpha)\bar{\theta}^i - \tilde{s}^i_t\big)^2 + \alpha_\theta(\bar{\theta}^i)^2\Big]dt.$$

FOC yields the closed form

$$\boxed{\bar{\theta}^i \;=\; \frac{\rho(1-\alpha)}{(1-\alpha)^2+\alpha_\theta}\int_0^\infty e^{-\rho t}\,\tilde{s}^i_t\,dt, \qquad \widehat{NFA}{}^i_0 \;=\; \frac{\alpha}{\rho}\,\bar{\theta}^i.}$$

Three determinants of the optimal transfer

Asymmetry

Transfers respond to $\tilde{s}^i = a_i - a^*$, the idiosyncratic shock; symmetric shocks $\Rightarrow$ no transfers.

Persistence

The transfer is proportional to the discounted cumulative shock, $\int_0^\infty e^{-\rho t}\tilde{s}^i_t dt$; more persistent shocks warrant larger transfers.

Openness (α)

The transfer is hump-shaped in openness: zero at $\alpha \in \{0, 1\}$, with an interior maximum.

- $\alpha \to 0$: small transfers but large expenditure-switching effects.
- $\alpha \to 1$: country too small to benefit.

Closed-economy limit

For exponential shocks $\tilde{s}^i_t = \tilde{s}^i_0 e^{-\psi t}$ and $\alpha \to 0$:

$$\widehat{NFA}^i_0 \;\to\; 0, \qquad \bar{\theta}^i \;\not\to\; 0.$$

- Net transfers vanish in dollar terms.
- The allocation differs significantly from the no-transfer case.
- Each unit of transfer has a large expenditure-switching effect.

Closed economies benefit ***more*** *per unit of transfer* – the opposite of McKinnon (1963).

Permanent shocks in the closed-economy limit

When the shock is permanent ($\psi = 0$):

$$\bar{y}_t^i = \bar{\pi}_{H,t}^i = 0, \qquad \bar{\theta}^i = \tilde{s}_0^i, \qquad \widehat{NFA}_0^i = 0.$$

The second-best attains the **first best**:

- Output and inflation gaps perfectly stabilized.
- Vanishingly small dollar transfers.
- Even rather closed economies in a currency union obtain large stabilization benefits with modest transfers.

Numerical illustration: calibration

Benchmark calibration (annual).

- Frisch elasticity $1/\varphi = 0.33, \quad \varphi = 3$
- Substitution elasticity across varieties $\varepsilon = 11$
- Discount rate $\rho = 0.06$
- Calvo arrival $\rho_\delta = 0.79$ (avg. price duration 5.6Q)
- Openness $\alpha \in \{0.01, 0.1, 0.4\}$
- Hand-to-mouth share $\chi \in \{0, 0.5\}$

Findings.

- Transfers strongly reduce gaps and inflation
- More effective in more closed economies
- More effective for more persistent shocks
- Hand-to-mouth consumers help for transitory shocks

Numerical illustration: stabilization gains

	$\alpha = 0.01$	$\alpha = 0.1$	$\alpha = 0.4$
Output gap, no transfer	-5.0%	-4.7%	-3.0%
Output gap, with transfer	$\approx 0\%$	$\approx -0.5\%$	$\approx -1.5\%$
Stabilization	$\approx 100\%$	$\approx 90\%$	$\approx 50\%$

- As the economy becomes more closed, optimal transfers deliver a larger fraction of the first-best stabilization.
- In the closed-economy limit, they attain it fully.

Takeaways

- **Currency unions and fiscal unions go hand in hand.** The loss of the exchange rate creates an aggregate-demand externality that only fiscal transfers can correct.
- **Fiscal $\neq$ financial integration.** Even with complete markets, private risk sharing is constrained inefficient. The two are complements, not substitutes.
- **Transfers target asymmetric shocks.** Monetary policy handles the symmetric component; $\bar{\theta}^i$ stabilizes the rest.
- **More persistent shocks $\Rightarrow$ larger transfers.** Effectiveness is hump-shaped in openness; closed economies benefit more per unit transferred.
- **Eurozone implication.** A systematic, rules-based mechanism (e.g. a common unemployment-insurance scheme) approximates the optimal design better than discretionary bailouts.

Thank you

Questions and discussion